Bio-IFGT: Biological Morphogenesis as Information-Flow Attractor Dynamics

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Abstract

This paper presents a minimal and unified framework for biological systems in terms of information density I, information flow $\mathbf{J}$, and an induced potential Φ. In this framework, DNA is treated as a parameter set θ that constrains system dynamics, rather than as an encoded final form.

On this basis, development is interpreted as convergence toward stable configurations, morphology as an attractor structure, and life as a dynamically sustained non-equilibrium state. The probability of the emergence of life is correspondingly reformulated as the probability of reaching a stable region in state space, rather than as the probability of hitting a single specified molecular configuration.

Intelligence is treated only as a later structure: a second attractor supported by the life attractor. The paper does not introduce a new biological mechanism. It provides a unified structural description of biological phenomena using a minimal set of dynamical quantities.

Contents

1 Introduction

Many biological phenomena are observed as development, morphogenesis, differentiation, and evolution. These phenomena are often described using language such as design, program, or genetic instruction. Such language can be useful at an intuitive level, but its relation to the underlying physical and chemical processes is not always explicit.

DNA sequences constrain molecular production and reaction pathways. They do not directly specify a completed structure, purpose, or intention. The fact that cells with the same DNA can differentiate into distinct states illustrates this point. DNA is therefore treated here as a parameter set θ that constrains the dynamics of the system.

This paper describes biological systems as processes of state change and sustained structure. We use three quantities: information density I, information flow $\mathbf{J}$, and induced potential Φ. These quantities provide a common description of development and morphogenesis.

The central feature of the framework is that it does not require design or purpose as explanatory primitives. Form and function appear as outcomes of state changes that converge toward stable structures.

The aim is to describe existing phenomena in a consistent form. We combine a minimal set of variables, a continuity equation, and an attractor-based interpretation of stability.

Section 2 gives the intuitive picture. Section 3 defines the formal mapping. Section 4 introduces the core equations. Sections 5 and following apply the framework to biological phenomena, simulation, and scope.

This paper builds on the Information-Field-Geometry-Theory (IFGT), which formulates the motion of systems toward closure and the persistence of quasi-closure structures. In IFGT, the degree of closure C characterizes how far a system has progressed toward a stable configuration. In biological systems, the field I should not be identified directly with the IFGT quantity $I = f(1-C)$. It is a biological, coarse-grained constraint-density field: a measure of local structural constraint, differentiation, and retained bias within a particular observational window. The relation is therefore a correspondence between descriptive layers, not an identity between variables. The sustained non-equilibrium structure of living systems corresponds, in IFGT terms, to quasi-closure: a regime that remains on the generated side of the complete-closure boundary and persists because residual difference continues to be available for update.

2 Intuitive Picture

This section gives the informal picture behind the formalism. The aim is not to define the framework rigorously, but to indicate what is being described.

2.1 State and Change

A biological system is not treated as a fixed object. It is treated as a process in which states continue to change. Molecules are produced, degraded, transported, and coupled through reactions. Cellular states change over time.

The important point is that change does not stop, while the system can still preserve an overall organization. Biological stability is therefore not static immobility. It is sustained organization under continuing update.

2.2 Bias and Flow

State change is not uniform. Some regions produce particular molecules more strongly. Other regions suppress them. These asymmetries give direction to state change. We call this directionality a flow.

The flow arises from production, diffusion, transport, and interaction. It is not a separate substance. It is the net direction and rate of state update.

2.3 Stable Regions

When flows exist, not all states behave in the same way. Some states are maintained more easily, while others decay or are displaced. As a result, regions can appear to which the system repeatedly returns.

Such regions are stable in the dynamical sense. They can absorb nearby deviations and preserve a recognizable structure.

2.4 Appearance of Form

Development and morphogenesis are interpreted as the appearance of stable configurations through continued state change. A form is not assumed in advance. It is observed when the dynamics settle into a stable arrangement.

Cellular distributions and organ-level patterns are described as such stable arrangements.

2.5 Without Design Assumptions

This view does not require purpose, design, or instruction as primitive concepts. It requires only state change, asymmetry, flow, and stable configurations.

2.6 Summary

The intuitive picture is that biological systems can be understood through state change, flow, and stable structure. The formal sections describe this picture using information density I, information flow $\mathbf{J}$, and induced potential Φ.

3 Formal Mapping

This section maps the intuitive picture onto formal quantities. The goal is to make the correspondence between intuition and equations explicit, not to derive a complete model for any particular organism.

3.1 State Description

A biological system is represented as a spatially and temporally varying state. For position $\mathbf{r} \in \mathbb{R}^3$ and time $t \in \mathbb{R}$, we consider coarse-grained fields rather than all microscopic molecular details.

3.2 Information Density

The bias or constraint present in a local state is represented by the information density

$$I(\mathbf{r}, t). \tag{1}$$

Here, information does not mean semantic content. It denotes the degree of local structure, constraint, differentiation, or interdependence.

3.3 Information Flow

The direction and rate of state update are represented by the information flow

$$\mathbf{J}(\mathbf{r}, t). \tag{2}$$

The flow combines contributions from reactions, diffusion, transport, and interactions. It describes how the coarse-grained state is updated.

3.4 Induced Potential

When information density is spatially biased, state change becomes asymmetric. This asymmetry is represented by an induced potential

$$\Phi = \Phi[I]. \tag{3}$$

This represents an effective potential induced by the distribution of I. The corresponding gradient-driven bias is written as

$$\mathbf{F} = -\nabla\Phi. \tag{4}$$

3.5 Relation to Flow

The information flow can be written as the sum of a diffusive component and a component induced by the potential:

$$\mathbf{J} = -D\nabla I + \mu\mathbf{F}, \tag{5}$$

where D is a diffusion coefficient and μ is a response coefficient.

3.6 Treatment of Time

The equations are written using an external time parameter t. This is the time with respect to which state change is recorded. We may also introduce an internal time τ that measures the progress of state update:

$$\frac{\mathrm{d}\tau}{\mathrm{d}t} = \rho(I, \mathbf{J}), \tag{6}$$

where ρ is a state-dependent progression rate. In this paper, equations of motion are written in t, while τ is used interpretively.

3.7 Summary

The mapping is

$$\text{bias or constraint} \mapsto I, \tag{7}$$

$$\text{state update} \mapsto \mathbf{J}, \tag{8}$$

$$\text{asymmetry in update} \mapsto \Phi. \tag{9}$$

With these quantities, the intuitive picture becomes a continuous-field description.

4 Core Equations

This section introduces the minimal equation system used throughout the paper. The equations are not tied to a specific biological system. They provide a general form for describing state change, flow, and stable structure.

4.1 Basic Variables

The basic fields are the information density

$$I(\mathbf{r}, t) \tag{10}$$

and the information flow

$$\mathbf{J}(\mathbf{r}, t). \tag{11}$$

The field I represents local density of structure or constraint. The field $\mathbf{J}$ represents the direction and rate of state change.

4.2 Continuity Equation

Information density obeys the continuity equation

$$\frac{\partial I}{\partial t} + \nabla \cdot \mathbf{J} = \sigma, \tag{12}$$

where $\sigma(\mathbf{r}, t)$ is a local generation rate. This equation records local production, outflow, and conservation-like transport of the coarse-grained quantity I.

4.3 Structure of the Flow

The flow is written as

$$\mathbf{J} = -D\nabla I + \mu \mathbf{F}, \tag{13}$$

where D is a diffusion coefficient, μ is a response coefficient, and $\mathbf{F}$ is an induced direction field.

4.4 Potential

The direction field is determined by the induced potential:

$$\mathbf{F} = -\nabla \Phi. \tag{14}$$

The potential is a functional of information density:

$$\Phi = \Phi[I]. \tag{15}$$

The specific form of Φ is system-dependent and left unspecified. Thus, biases in information density induce a gradient-driven bias in state change. The sign convention is fixed only after $\Phi[I]$ is specified. For example, choosing $\Phi \sim -I$ makes the induced component $-\nabla\Phi$ point toward increasing constraint density. Other biological observables may require the opposite convention or additional response terms.

4.5 Dynamical Loop

Equations (12)–(15) form a loop: the distribution of I induces Φ, the potential shapes $\mathbf{J}$, and the flow redistributes I. In general this loop does not necessarily terminate in a static equilibrium. It can produce non-equilibrium structures that are sustained under continued update.

4.6 Two-Layer Description of Time

The equations use the external time parameter t. An internal time τ can be introduced as

$$\frac{\mathrm{d}\tau}{\mathrm{d}t} = \rho(I, \mathbf{J}), \tag{16}$$

where $\rho(I, \mathbf{J})$ is a state-dependent progression rate. The variable t is used for equations of motion. The variable τ is used to interpret the internal progress of the system.

4.7 Stable Structures

Under these dynamics, a region of state space is stable if states within or near it tend to remain there or return after perturbation. Such a region is interpreted as an attractor.

4.8 Summary

The minimal equation system is

$$\begin{align}
\frac{\partial I}{\partial t} + \nabla \cdot \mathbf{J} &= \sigma, \tag{17}\\
\mathbf{J} &= -D\nabla I + \mu \mathbf{F}, \tag{18}\\
\mathbf{F} &= -\nabla \Phi, \tag{19}\\
\Phi &= \Phi[I]. \tag{20}
\end{align}$$

This system describes state change, flow formation, and the emergence of stable structures in a common form.

5 Gene × Information Flow

This section describes DNA and gene-related processes in terms of information density and information flow. The aim is to describe the emergence of development, form, and cell fate as sequences of state change, rather than as the execution of semantic instructions.

5.1 Role of DNA

A DNA sequence s is written as

$$s \in \{A, T, G, C\}^N. \tag{21}$$

The sequence has no intrinsic meaning or purpose in the present formalism. It constrains molecular possibilities, reaction propensities, and threshold conditions.

Thus DNA is treated as a parameter set

$$\theta = \theta(s), \tag{22}$$

which constrains system dynamics. It is not treated as a direct specification of a completed form.

5.2 Gene Events as State Changes

Transcription, translation, binding, and modification are represented as local operations that change information density:

$$I(\mathbf{r}, t) \mapsto I(\mathbf{r}, t) + \Delta I. \tag{23}$$

Although these events are discrete, their aggregate effect can be represented as a continuous information flow $\mathbf{J}(\mathbf{r}, t)$. A gene network is therefore interpreted as a relational structure that generates information flow.

5.3 Connection to the Continuity Equation

In the continuity equation

$$\frac{\partial I}{\partial t} + \nabla \cdot \mathbf{J} = \sigma, \tag{24}$$

the source term σ represents information generation due to gene-related events. The flow $\mathbf{J}$ represents the accumulated effect of such events across space and time.

5.4 Bias and Induced Potential

When information density becomes biased, it induces a potential

$$\Phi = \Phi[I]. \tag{25}$$

State change is then biased according to

$$\mathbf{F} = -\nabla \Phi. \tag{26}$$

This is not a new fundamental force. It describes the ordinary dynamical fact that stable states are more easily maintained and can draw nearby states into their basin.

5.5 If–Then Conditions as Thresholds

The conditional effects associated with DNA are implemented as threshold conditions in a continuous field. For a response function F_k, a reaction may switch when

$$F_k(I, \mathbf{J}, \mathbf{r}, t; \theta) > \theta_k. \tag{27}$$

Here θ_k is a parameter induced by the DNA-dependent parameter set θ. The condition is not a symbolic rule in isolation. It is a threshold in the dynamics.

5.6 Interpretation of Form

The resulting picture is as follows. DNA supplies parameters. Information density and information flow are formed. Biases in I induce Φ. Flow is shaped by the induced potential. Thresholds branch the state. Stable configurations then appear as attractors. Cell fate and morphology are observed as such stable configurations.

5.7 Scale Invariance

The same formal structure can be used at molecular, cellular, and tissue scales. The level of coarse-graining changes, but the variables I, $\mathbf{J}$, Φ, thresholds, and stable solutions retain the same roles.

5.8 Summary

DNA is treated as a condition-setting parameter set. Biological structure appears as a stable solution formed by information density and information flow. This provides the basis for the treatment of development and morphogenesis in the next section.

5.9 Origin of the Parameter Set

At this descriptive layer, the parameter set θ is not treated as a primitive. In IFGT terms, θ may be read as a quasi-closure structure stabilized through prior dynamics of difference: a configuration retained by the persistence of certain closure patterns over others. DNA sequences are therefore interpreted here as historically stabilized constraint patterns that persisted because they supported sustained non-equilibrium dynamics. This does not derive the concrete origin of θ for a given organism. It only states that θ need not be placed outside the framework as an explanatory object.

6 Development and Morphogenesis

This section applies the preceding framework to development and morphogenesis. Development is interpreted not as execution of a design, but as a process in which state changes converge toward stable structures.

6.1 Initial State and Asymmetry

A fertilized egg is an initial state with broad potential in state space. At this stage, information density may be approximately uniform, and no macroscopic structure is yet fixed.

Small asymmetries nevertheless exist through initial conditions and environmental fluctuations. These asymmetries can seed later structure formation.

6.2 Morphogens as Information Density

During development, morphogens form spatial concentration profiles. In the present framework, such profiles are represented as information density:

$$I(\mathbf{r}, t). \tag{28}$$

Local production creates gradients in I. Diffusion and degradation then shape the resulting distribution.

6.3 Dynamics

A standard coarse-grained form for this process is

$$\frac{\partial I}{\partial t} = D\nabla^2 I - \lambda I + \sigma(\mathbf{r}, t; \theta), \tag{29}$$

where D is a diffusion coefficient, λ is a decay rate, and σ is a local source constrained by the parameter set θ.

This equation is not special to this framework. It is a standard reaction–diffusion form used here as a component of the information-flow description.

6.4 Induced Potential and Gradient

The distribution of I induces a potential

$$\Phi = \Phi[I]. \tag{30}$$

State change is biased by

$$\mathbf{F} = -\nabla\Phi. \tag{31}$$

As a result, states become biased toward regions or gradients associated with I. This bias gives directionality to structure formation.

6.5 Differentiation by Thresholds

Cell fate can branch according to thresholds in I. For example,

$$I(\mathbf{r}) < \theta_1 \quad \Rightarrow \quad \text{state A}, \tag{32}$$
$$\theta_1 \leq I(\mathbf{r}) < \theta_2 \quad \Rightarrow \quad \text{state B}, \tag{33}$$
$$I(\mathbf{r}) \geq \theta_2 \quad \Rightarrow \quad \text{state C}. \tag{34}$$

The variable I is continuous, while the observed outcome may be discrete. Differentiation is therefore interpreted as a threshold response to continuous state change.

6.6 Morphology as Attractor

The combination of dynamics and thresholds stabilizes distributions and fixes differentiation patterns. The stable state is the morphology observed in development. Morphology is thus not an independently specified object. It is the result of dynamical convergence.

6.7 Robustness

Development is often reproducible and robust. In this framework, robustness follows from attractor structure: multiple nearby initial states can converge to the same stable configuration. Local fluctuations are absorbed by the dynamics.

6.8 Summary

Development and morphogenesis are described through small initial asymmetries, distributions of information density, gradient-driven bias, threshold branching, and convergence to attractors. No design or purpose is required as an additional explanatory primitive.

7 Life as Attractor

This section formulates life not as a completed object, but as a sustained stable structure in state space. This view allows maintenance, development, differentiation, and regeneration to be described within one dynamical framework.

7.1 State-Space Description

A biological system is represented by a state vector

$$x(t) = (x_1(t), x_2(t), \ldots, x_n(t)), \tag{35}$$

where the components may include molecular concentrations, expression levels, reaction rates, and local environmental variables.

The system evolves as

$$\frac{\mathrm{d}x}{\mathrm{d}t} = F(x; \theta), \tag{36}$$

where θ is the parameter set associated with DNA-dependent constraints such as reaction propensities and thresholds.

7.2 Internal Time

In addition to external time t, we introduce an internal time τ that measures the progress of state update:

$$\frac{\mathrm{d}\tau}{\mathrm{d}t} = \rho(x). \tag{37}$$

The function $\rho(x)$ is a state-dependent progression rate. The variable τ represents accumulated state change or information update, and need not coincide with t. Equations of motion are written in t; τ is used to interpret their internal progression.

7.3 Definition of Life

Life is not a single state. It is motion that continues while remaining within a bounded region of state space. Such motion is not static, does not diverge arbitrarily, and remains constrained within a viable range. In this regime, τ continues to increase.

7.4 Life as Attractor

Let Ω be the state space and let

$$\Omega_{\text{life}} \subset \Omega \tag{38}$$

be a region such that, once the system enters it, the dynamics tend to return to it after moderate perturbation. We call Ω_{life} the life attractor.

This region is not externally imposed. It is formed through interactions within the system and its environment.

7.5 Non-Equilibrium Sustained Structure

The life attractor is not an equilibrium state. Matter enters and leaves, molecules are replaced, and energy is consumed. Nevertheless, the organization persists. Life is therefore interpreted as a sustained structure in a non-equilibrium state.

7.6 Relation to Development

The developmental process described in Section 6 can be interpreted as convergence from an initial state toward the life attractor. Reproducibility and robustness arise when multiple initial conditions converge to the same attractor region.

7.7 Differentiation and Sub-Attractors

Within the life attractor, more specific stable regions may exist. These are sub-attractors. Cell differentiation is interpreted as transition into such sub-attractors within Ω_{life}. This structure can describe discreteness, plasticity, and partial reversibility without treating differentiated states as externally specified endpoints.

7.8 Robustness and Regeneration

Robustness follows from the attractor property. When perturbations displace the system, the dynamics can return it toward the stable region. Regeneration is interpreted as a case in which attractor structure is reconstructed after a larger disturbance.

7.9 Summary

Life is not merely a collection of matter or a static structure. It is dynamically sustained stable motion in state space. This motion is described through dynamics involving information density I, information flow $\mathbf{J}$, and induced potential Φ. The next section uses this view to revise the formulation of emergence probability.

8 Probability Revision

This section revisits the formulation of the probability of the emergence of life. The aim is not to assign a numerical value, but to clarify what the probability is taken to measure.

8.1 Conventional Formulation

The emergence of life is often treated as the probability that a particular molecular structure appears by chance. In this view, many conditions must be satisfied simultaneously, and each may have low probability. The combined probability can therefore become very small.

This view is coherent if life is assumed to be a particular completed configuration. The present framework uses a different object of probability.

8.2 The Issue

As described in Section 7, life is not a fixed structure. It is sustained stable motion in state space. Life therefore corresponds not to a single point, but to a region with finite extent.

The emergence of life is then interpreted as reaching this region and remaining within it, rather than hitting one specified structure.

8.3 Probability as Region Entry

Let the life attractor occupy a region

$$\Omega_{\text{life}} \subset \Omega. \tag{39}$$

The probability of life emergence is then the probability of reaching Ω_{life}. This is a probability of region entry, not a probability of exact point matching.

8.4 Structure of Trials

Molecular reactions and material cycles repeat as long as enabling conditions persist. States that do not sustain themselves disappear. States that sustain themselves remain. In this sense, life appears not because it is selected in advance, but because certain states can persist.

8.5 Dimensional Reduction

In a point-target view, many degrees of freedom must be specified simultaneously. In the present framework, the relevant conditions for life are described by fewer effective variables that characterize entry into a stable region. The dimensional structure of the probability problem is therefore changed.

8.6 Interpretation

This is not a numerical reassessment of probability. It is a structural redefinition of the object being counted. Life is neither a miraculous hit nor an automatic necessity. It is a state that can appear when sustaining conditions exist.

In IFGT terms, this reformulation has a precise interpretation: life corresponds to the entry and persistence of a system within a quasi-closure regime, a dynamical region in which the drive toward closure is sustained while complete closure remains unattained as an ordinary dynamical state. The probability of life is therefore better understood as the probability of reaching and maintaining such a regime, rather than the probability of assembling a fixed molecular configuration. The apparent improbability of life may reflect, in part, a consequence of how the target was identified.

8.7 Connection to Intelligence

This reformulation makes the next question more precise: after life is established, why do only some systems reach intelligence? Intelligence is not a simple continuation of life. It requires additional structural conditions. The next section treats intelligence as a second attractor.

9 Intelligence as a Second Attractor

This section treats intelligence not as a direct extension of life, but as a second stable structure supported by the life attractor. This view separates the conditions for life from the conditions for intelligence.

9.1 Position of Intelligence

Intelligence is often treated as an advanced function of living systems. However, life is already an attractor structure. Many living systems persist for long periods within that attractor without developing intelligence. Intelligence is therefore not merely a strengthening of the life attractor.

9.2 Second Attractor

Intelligence appears when information is not only flowing, but also retained, referenced, and reused. Past states influence present states, and present states modify future trajectories. This creates recurrent loops of information flow.

Such loops stabilize under conditions that differ from those required for life alone. Intelligence is therefore interpreted as a second attractor supported by, but not identical to, the life attractor.

9.3 Structural Independence

Formally, one may write

$$\Omega_{\text{intel}} \subset \Omega_{\text{life}}, \tag{40}$$

but the internal structure of Ω_{intel} is determined by additional constraints. These include persistent reference to state history, temporal accumulation of information, and sharing of information across multiple components. Such conditions are not required for biological maintenance alone.

9.4 Probability Structure

Following Section 8, the emergence of intelligence is treated as the satisfaction of conditions distinct from the emergence of life. Its probability therefore has a conditional structure. A schematic factorization is

$$P_{\text{intel}} \approx P_{\text{life}}\, P_{\text{second}|\text{life}}, \tag{41}$$

where $P_{\text{second}|\text{life}}$ denotes the probability of reaching the second attractor given the life attractor. This expression is not a numerical estimate. It records the two-stage structure of the problem.

9.5 Non-Teleological Interpretation

Treating intelligence as a second attractor does not give it a privileged status. An attractor is not a goal. It is a structure that persists when its conditions are satisfied. Intelligence is therefore neither inevitable nor a final form. It is one possible stable structure.

9.6 Relation to Internal Time

In the notation of Section 7, life is a structure in which τ continues to progress. Intelligence is characterized by the reference and reuse of the history of τ. In this sense, intelligence is not merely flow. It is recursive information flow.

9.7 Summary

Intelligence is not treated as a simple extension of life. It is a stable structure of recursive information flow supported by the life attractor. This interpretation explains why intelligence has additional conditions and is not universal among living systems.

9.8 Connection to Further Work

The next question is how this second attractor is implemented. Its realization may involve memory, temporal integration, language, and inter-individual sharing. These issues are beyond the minimal biological framework developed here.

10 Relation to Note

This section clarifies the relation between this paper and the corresponding note-style exposition. The aim is not to summarize the theory, but to explain how the same structure appears in different descriptive layers.

10.1 Difference in Descriptive Layer

This paper is written as an L3 formal layer. The note essay is written as an L1 intuitive layer. They concern the same target, but their purposes differ. The paper fixes definitions, equations, and relations. The note essay supports intuitive understanding through accessible language and metaphor.

The relevant relation is therefore structural correspondence, not literal identity of wording.

10.2 Correspondences

The concepts used in this paper correspond to intuitive expressions in the note essay:

$$I \leftrightarrow \text{local concentration, bias, or structured region,} \tag{42}$$

$$\mathbf{J} \leftrightarrow \text{flow, repetition, or ease of passage,} \tag{43}$$

$$\Phi \leftrightarrow \text{return tendency or stable place,} \tag{44}$$

$$\text{attractor} \leftrightarrow \text{returning to the same region or preserving form.} \tag{45}$$

These are not direct translations. They are expressions of the same phenomenon at different resolution.

10.3 Development and Form

The treatment of development and morphogenesis in Section 6 corresponds to the note-level idea that repeated changes form flows and that flows stabilize into form. Both descriptions avoid taking design or purpose as primitives.

10.4 Definition of Life

Section 7 defines life as sustained stable motion. In the note essay, this appears as the idea that life is not an object but a state. The two expressions refer to the same structure at different levels of precision.

10.5 Probability Revision

The probability revision in Section 8 corresponds to the note-level statement that life is not best understood as a miracle-like exact hit. The shared structure is the shift from a point target to a region, and from exact matching to arrival and persistence.

10.6 Connection to Intelligence

The treatment of intelligence as a second attractor in Section 9 corresponds to a note-level description of a flow that appears to think. The note essay does not specify the full structure, but it preserves the distinction between life and intelligence as different stable organizations.

10.7 Relation to Appendices

Intermediate appendices can bridge the note essay and this paper. They may introduce state, flow, and stability with minimal mathematics. Future versions can be updated to reflect the structure fixed here.

10.8 Summary

This paper and the note essay differ by descriptive form, not by subject. The paper fixes the formal structure. The note essay presents the same structure experientially. Each can be read independently, but the correspondence improves coherence across layers.

11 Minimal Developmental Simulation (Bio-IFGT Toy Model)

11.1 Purpose

This section introduces a minimal Bio-IFGT toy model. The purpose is to illustrate how a small set of DNA-like parameters, including sources, diffusion, decay, thresholds, and response rates, can generate stable biological-like structures without explicitly encoding their geometry.

11.2 Model Setup

We consider a two-dimensional domain representing an embryo-like field. The system evolves over discrete time steps and consists of four primary fields:

- morphogen field $M(x, y, t)$,
- information density $I(x, y, t)$,
- neural commitment $N(x, y, t)$,
- eye-like commitment $E(x, y, t)$.

An auxiliary boundary field is also introduced:

- boundary stabilization $B(x, y, t)$.

11.3 Dynamics

Morphogen field. The morphogen field evolves by diffusion, decay, and localized production:

$$\frac{\partial M}{\partial t} = D_M \nabla^2 M - \lambda_M M + \sigma_M(x, y). \tag{46}$$

Information density. The information density accumulates from the morphogen field while also diffusing and decaying:

$$\frac{\partial I}{\partial t} = D_I \nabla^2 I - \lambda_I I + \alpha M. \tag{47}$$

Neural commitment. Neural commitment is updated by thresholding the information density:

$$N_{t+1} = N_t + \eta_N S(I - \theta_N)(1 - N_t). \tag{48}$$

Boundary field. The boundary field is derived from the gradient structure of information density:

$$G = |\nabla I|. \tag{49}$$

The boundary field represents regions where spatial variation is locally maximized. It then stabilizes through a thresholded update:

$$B_{t+1} = B_t + \eta_B S(G - \theta_B)(1 - B_t). \tag{50}$$

Eye-like commitment. The eye-like commitment field is updated only where neural commitment, boundary stabilization, and a positional window are simultaneously active:

$$E_{t+1} = E_t + \eta_E S(N - \theta_{EN}) S(B - \theta_{EB}) A(x, y)(1 - E_t). \tag{51}$$

The thresholding function is

$$S(z) = \frac{1}{1 + e^{-kz}}. \tag{52}$$

11.4 Interpretation

The model encodes no explicit geometric description of neural or eye-like structures. Instead, it specifies localized sources σ_M, diffusion and decay parameters, threshold conditions, and response rates. From these conditions, a hierarchical structure appears:

1. a smooth morphogen gradient forms in M,
2. an information-density landscape develops in I,
3. a neural-plate-like region emerges through thresholding in N,
4. boundary structure arises from gradients in B,
5. eye-like localized structures appear through secondary thresholding in E.

11.5 Key Observation

The emergence proceeds through successive constraints:

$$M \to I \to N \to B \to E. \tag{53}$$

Each stage introduces a higher-order structural condition:

- I: scalar field,
- ∇I: first-order structure,
- thresholding: discretization,
- combined conditions: localization.

11.6 Conceptual Implication

The simulation illustrates that biological-like structure need not be encoded as a completed form. It can arise as a stable solution within a constrained information landscape.

In this sense, DNA-like parameters act as conditions that shape the geometry of possible states.

11.7 Relation to Bio-IFGT

Within the Bio-IFGT framework:

- I corresponds to information density,
- $\Phi \sim -I$ defines an effective potential landscape,
- $\mathbf{J} \sim -\nabla\Phi$ induces flows toward stable regions.

The observed structures correspond to attractor-like configurations of this field.

Each stage in the generation path $M \to I \to N \to B \to E$ can be interpreted as a successive localization of quasi-closure: the morphogen field establishes a global information gradient, which drives local accumulation of I, which in turn crosses thresholds that stabilize sub-regions of the field. The eye-like commitment E represents a secondary quasi-closure, supported by but distinct from the neural-plate-like region N. This hierarchical structure mirrors the attractor-within-attractor interpretation of differentiation developed in Section 7.7.

11.8 Limitations

This model is intentionally minimal and does not attempt to reproduce biological detail. It serves only to demonstrate the structural principle of emergence under constrained dynamics.

12 Simulation Results and Visualization

This section reports a numerical run of the minimal developmental model introduced in the previous section. The aim is to show how structure appears over time in the toy system. The result is not intended as a reproduction of a specific biological process.

12.1 Initial State

At the initial time, the fields are nearly uniform and no organized structure is present. The morphogen field M has only a localized source term. The biological constraint-density field I is low and has no meaningful spatial bias apart from a small perturbation. The neural commitment field N and the eye-like commitment field E are initially undifferentiated.

In this state, no information corresponding to a completed shape exists anywhere in the field.

12.2 Time Evolution

As the system evolves, structure appears in the following order:

1. the morphogen field diffuses and forms a smooth gradient,
2. the biological constraint-density field I accumulates and develops spatial bias,
3. thresholding produces a central neural-plate-like region in N,
4. the gradient magnitude $|\nabla I|$ defines a boundary field B,
5. the overlap of neural commitment, boundary stabilization, and a positional window produces localized eye-like commitment in E.

This process is continuous. No stage contains an encoded final drawing of the resulting structures.

12.3 Final State

In the final state, the simulation shows a stable central neural-plate-like region. Localized eye-like structures appear near boundary regions of the neural field. The central region itself does not produce the eye-like field, because the relevant boundary condition is weak there.

Thus, the dynamics determine both where a structure can appear and where it does not appear.

12.4 Figure Descriptions

Each figure corresponds to a stage in the hierarchical emergence described in the previous section.

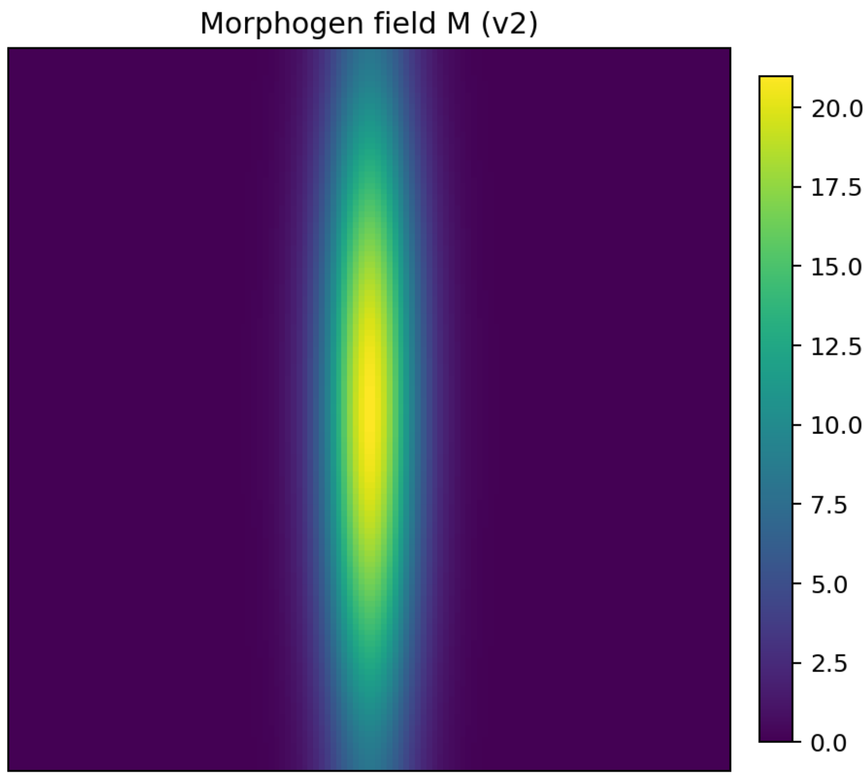

Figure 1: Final morphogen field M. A smooth field forms around the central source.

12.5 Time-Evolution Visualization

The time evolution was also rendered as an animated GIF:

`simulations/minimal_neural_field/outputs/development_v2_hierarchical.gif`

Time evolution of the minimal developmental model. Structures appear as stable configurations under simple dynamical constraints. The animation is provided as supplementary material. The animation shows the sequence of morphogen formation, biological constraint-density accumulation, neural-plate-like stabilization, boundary extraction, and secondary eye-like commitment.

This ordering agrees with the hierarchical generation path

$$M \to I \to N \to B \to E. \tag{54}$$

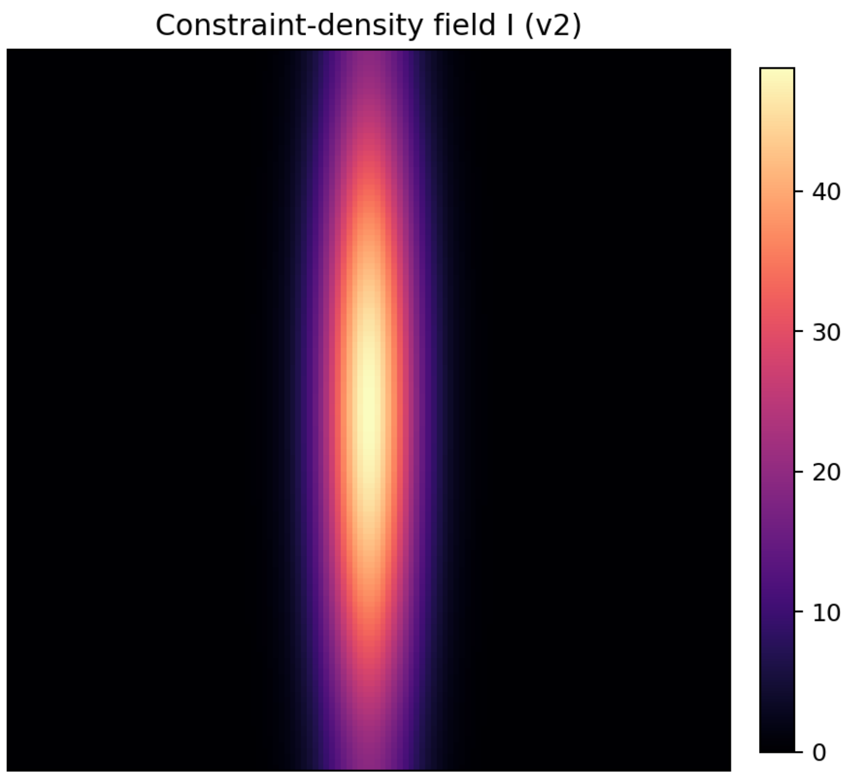

Figure 2: Final biological constraint-density field I. The field accumulates in response to the morphogen field and forms a central peak.

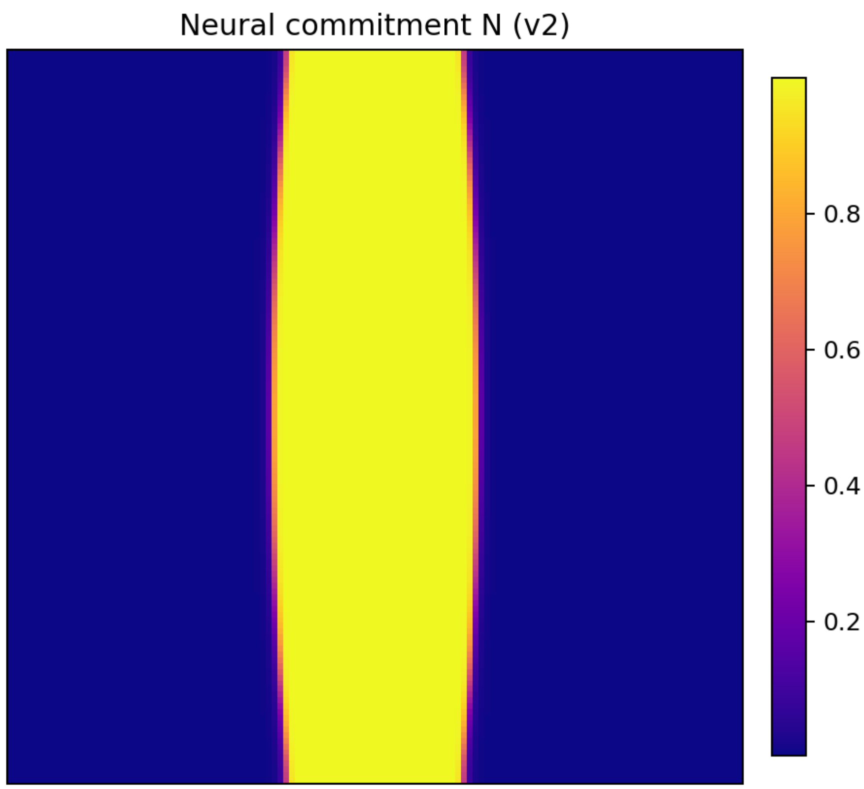

Figure 3: Final neural commitment field N. Thresholding of biological constraint density produces a central neural-plate-like region.

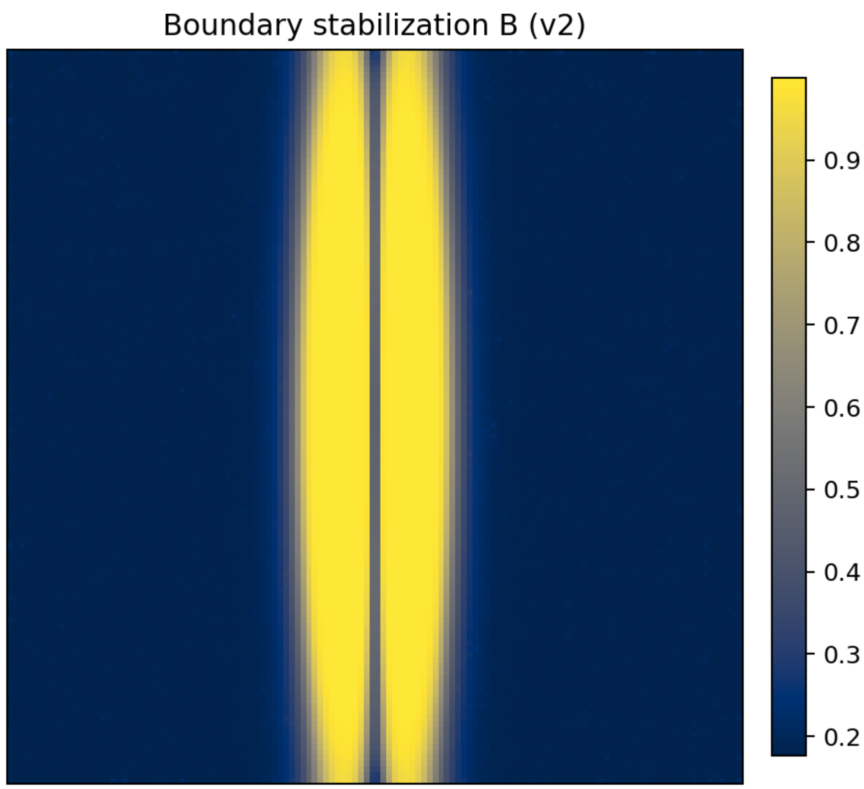

Figure 4: Final boundary stabilization field B. The field is derived from the gradient structure of I, highlighting boundary-like regions.

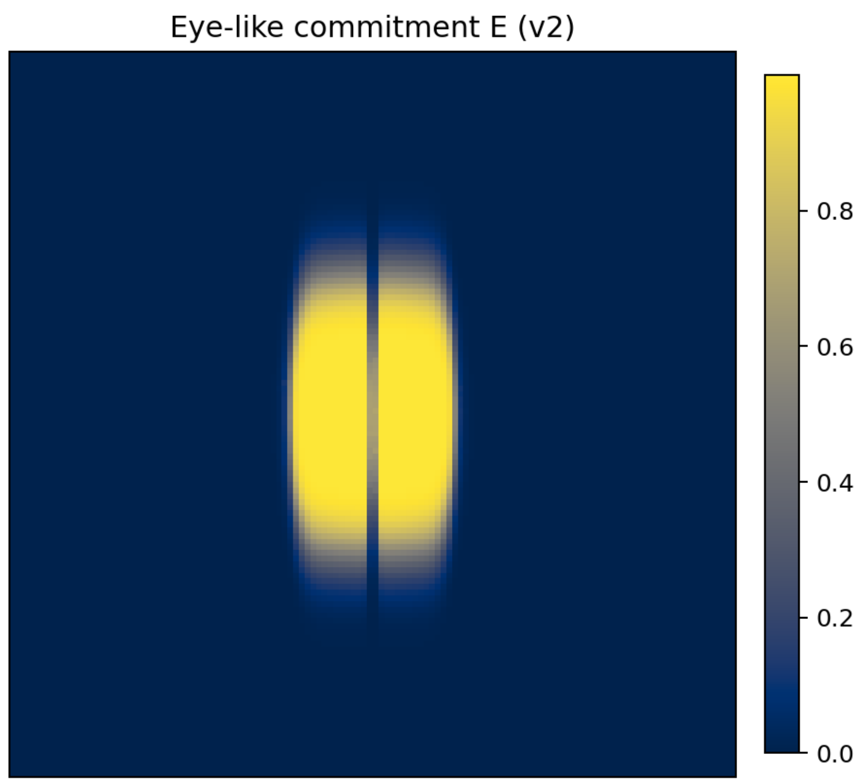

Figure 5: Final eye-like commitment field E. Localized structures appear where neural commitment and boundary conditions overlap within the positional window.

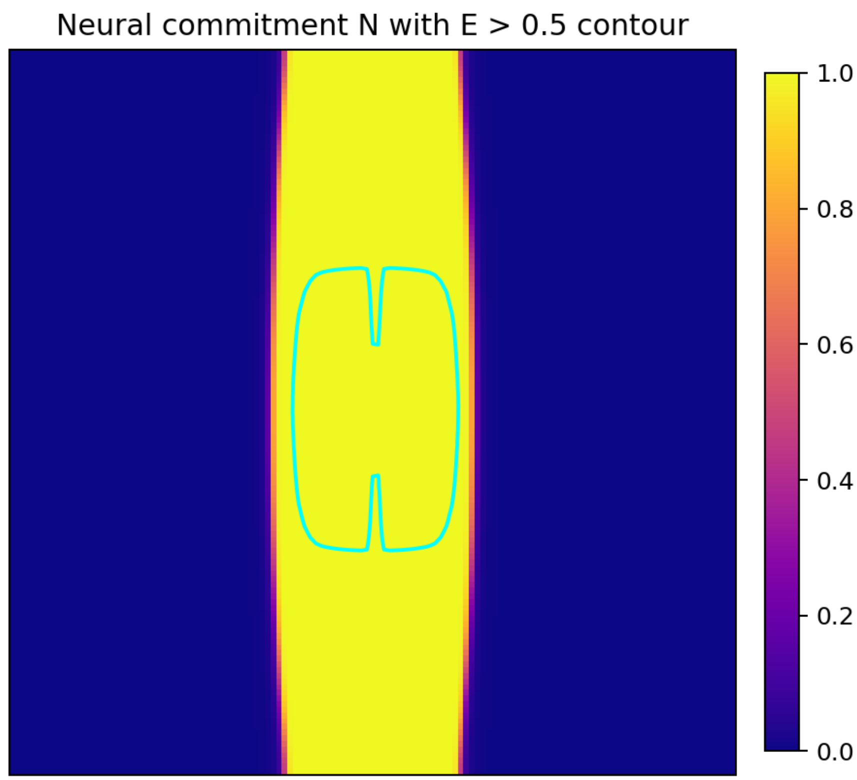

Figure 6: Overlay visualization. The neural commitment field N is shown as the background, with the thresholded eye-like commitment E overlaid as a contour.

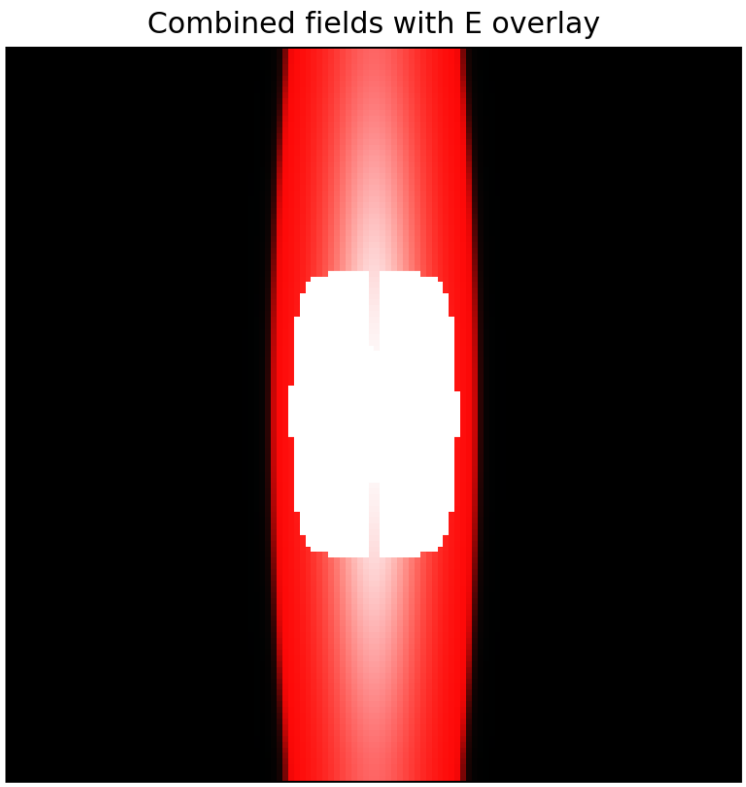

Figure 7: Combined visualization with eye-like commitment. The image summarizes the final relationship among the morphogen field, biological constraint density, neural commitment, and eye-like commitment.

12.6 Summary of Observations

The simulation illustrates the following points:

- the neural-plate-like region arises through thresholding of biological constraint density,
- the eye-like field arises through the combination of boundary structure and positional conditions,
- structure appears in stages,
- the final shape is not specified in advance.

The resulting structures are therefore interpreted as stable solutions under constrained dynamics.

12.7 Role of This Section

This result is not a model of a specific biological phenomenon. The role of the simulation is to show that structure can arise from minimal conditions in a field-based dynamical system. In this respect, the toy model is consistent with the Bio-IFGT interpretation developed in the preceding sections.

It does not demonstrate biological mechanism, but only structural possibility.

13 Discussion

This section summarizes the scope and limits of the framework. The aim is not to add new claims, but to clarify what the framework does and does not address.

13.1 Scope of Description

The paper describes biological systems using information density I, information flow $\mathbf{J}$, and induced potential Φ. This is a minimal language for treating development, morphogenesis, life maintenance, and the conditions for intelligence in a common form.

The framework extracts structure. It does not attempt detailed reproduction of each individual biological phenomenon.

13.2 Relation to Existing Models

The framework does not replace existing biological models. Reaction–diffusion systems, gene networks, and neural dynamics remain necessary at their own levels. The present formalism provides a way to redescribe such models in terms of common structural elements: density, flow, induced bias, and stability.

It does not require the assumption of a new mechanism, nor does it deny existing theories.

13.3 Non-Closure and Sustained Dynamics

The dynamics considered here do not generally converge to complete equilibrium or fixed closure. Flow continues, and structure is updated. In this sense, biological systems are interpreted as sustained non-closed structures.

This property is not developed as a separate theory in this paper. It appears as a descriptive feature of the dynamics considered.

This interpretation is consistent with the quasi-closure structure formalized in IFGT: biological systems remain within regimes where closure is partial, dynamically maintained, and not realized as an ordinary complete-closure state. The non-closure of biological dynamics is therefore not a limitation of the framework, but its central descriptive feature.

13.4 Status of Time

The paper uses external time t and introduces internal time τ as an interpretive variable. This separates observed time from the progress of internal state update. A precise operational definition of τ is outside the scope of the present paper.

13.5 Limits

Several limits should be stated. The framework does not derive the parameter set θ for a concrete organism. The explicit forms of I, $\mathbf{J}$, and Φ are system-dependent. The paper does not provide numerical predictions.

The framework is therefore a unifying description, not a detailed model of a particular biological system.

13.6 Future Directions

Future work should identify I, $\mathbf{J}$, and Φ in concrete systems, analyze the structure of parameter spaces, and quantify attractor regions. For intelligence, further work must address memory, temporal integration, language, and interaction among individuals.

13.7 Summary

The framework describes biological phenomena in a single formal language without using design or purpose as primitives. It does not address all mechanistic detail or provide numerical prediction. It is best understood as a basis for description and later specialization.

14 Conclusion

This paper described a minimal framework for biological systems using information density I, information flow $\mathbf{J}$, and induced potential Φ. Within this framework, development is interpreted as convergence toward stable structure, life as sustained stable motion, and intelligence as a second attractor supported by life.

The main point is that biological form can be described without treating design or purpose as primitive explanatory terms. The paper does not aim to reproduce every biological mechanism in detail. It extracts a common structure across biological phenomena.

The framework uses a small set of elements: state, flow, induced bias, and stability. These elements allow development, morphogenesis, life maintenance, probability revision, and the later emergence of intelligence to be placed in one descriptive form.

Several questions remain open. The parameter set θ must be specified for concrete systems. The empirical forms of I, $\mathbf{J}$, and Φ must be identified. The implementation of intelligence requires additional modeling.

The result of the paper is therefore not intended as a final theory. It is a form of description. Its usefulness depends on future applications and refinements.